

Overview and Comparison of Gaussian Process-Based Surrogate Models for Mixed Continuous and Discrete Variables: Application on Aerospace Design Problems



Julien Pelamatti, Loïc Brevault, Mathieu Balesdent, El-Ghazali Talbi and Yannick Guerin

Abstract Surrogate modeling is an increasingly popular tool for engineering design as it enables to model the performance of very complex systems with a limited computational cost. A large number of techniques exists for the surrogate modeling of continuous functions, however, only a very few methods for the surrogate modeling of mixed continuous/discrete functions have been developed. In this chapter, existing adaptations and variants of Gaussian process-based surrogate modeling techniques for mixed continuous/discrete variables are described, discussed and compared on several analytical test-cases and aerospace design problems.

1 Introduction

Nowadays, because of cost and scheduling constraints, a large part of engineering design projects relies on computer models and simulations, as it is usually a cheaper and fairly accurate alternative to physical testing. However, computer modeling also

J. Pelamatti

ONERA/DTIS, Université Paris Saclaym, CNES, Direction des lanceurs,
University of Lille, Lille, France
e-mail: julien.pelamatti@onera.fr

L. Brevault · M. Balesdent

ONERA/DTIS, Université Paris Saclay, 91123 Palaiseau Cedex, France
e-mail: loic.brevault@onera.fr

M. Balesdent

e-mail: mathieu.balesdent@onera.fr

E.-G. Talbi (✉)

Polytech Lille – University of Lille CNRS/CRISAL, Inria Lille – Nord Europe,
59650 Villeneuve d’Ascq, Lille, France
e-mail: el-ghazali.talbi@univ-lille.fr

Y. Guerin

CNES, Direction des lanceurs, 75612 Paris Cedex, Paris, France
e-mail: yannick.guerin@cnes.fr

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T. Bartz-Beielstein et al. (eds.), *High-Performance Simulation-Based Optimization*, Studies in Computational Intelligence 833,
https://doi.org/10.1007/978-3-030-18764-4_9

presents a number of limitations and drawbacks. Most notably, the computation time for complex system simulations can become considerably large, as for instance occurs when finite element calculations or computational fluid-dynamics are involved. This issue becomes even more problematic when performing system optimizations, as the computer models must be called a large amount of times. In order to avoid this problem, it is common practice to create a surrogate of the numerical model [31], also known as surrogate model, which relies on a simple analytical representation of the studied function, resulting in much shorter computation time. However, the addition of a modeling layer implies a loss of accuracy, the magnitude of which depends on the type of modeling technique that is used as well as on its parametrization.

Most of the existing surrogate modeling techniques have been developed for functions which depend solely on continuous variables. However, complex system performance are often modeled by functions which depend simultaneously on both continuous and discrete variables. In these cases, a number of additional challenges need to be addressed in order to be able to create an accurate and reliable surrogate model. A common example of such a design problem is the preliminary design of launch vehicles, which involves discrete architectural and technological choices (e.g., type of propulsion, number of stages, type of material) for which a judicious combination determines the pertinence and the performance of a new configuration. In the recent years, a few adaptations of existing continuous surrogate modeling techniques as well as novel modeling methods have been developed in order to be able to model mixed continuous/discrete functions. A number of reviews of existing mixed continuous/discrete surrogate modeling techniques exists [3, 26, 39], in which 3 separate approaches can be identified. The first one consists in converting the discrete variables into continuous one through a so-called ‘coding’ technique, such as direct conversion [20], dummy coding [18], effect coding [40] and contrast coding [9], and subsequently including them in standard continuous surrogate models such as Support Vector Machine [18] or moving least square regression [11]. A second approach to the surrogate modeling of mixed continuous/discrete functions consists in using a definition of extended distance between two samples adapted to mixed search spaces. This approach has been applied to modeling methods such as Gaussian processes [16] and Radial Basis Functions [22]. Finally, several surrogate modeling methods were also developed for mixed continuous/discrete functions by adapting their working principle to the presence of discrete variables. Notable examples of these approaches are the Categorical Regression Spline [28] and the hypersphere decomposition based Gaussian process [10, 41]. To the best of the authors’ knowledge, no comprehensive review of existing Gaussian process-based surrogate models for mixed continuous/discrete variables, such as the one provided in this chapter, exists. Furthermore, in the existing mixed variable surrogate model reviews the comparison are only performed on standard analytical benchmarks, while in this chapter “Multi-objective Bayesian Optimization for Engineering Simulation” distinct representative aerospace engineering related simulations are used for the bench-marking of the presented techniques.

This chapter focuses on existing Gaussian process based modeling techniques for mixed continuous/discrete functions, as they provide the user with a prediction of the

modeled function value as well as an estimate of the corresponding surrogate modeling uncertainty. This characteristic is particularly useful when performing surrogate model based optimization (such as Efficient Global Optimization [19]), as it enables to refine the model in an efficient fashion. Following this introduction, the second section provides a brief theoretical description of the mixed continuous/discrete modeling problem and all the related challenges, followed by an overview of the existing approaches to these problems. In the third section, Gaussian process-based surrogate modeling is described in details and its existing adaptations for the modeling of mixed continuous/discrete functions are analyzed, discussed and compared. In the fourth section, the application of these Gaussian process techniques performed on different test-cases is discussed. First, three analytical test-cases of varying dimension and complexity are presented, allowing to better illustrate the various surrogate modeling techniques and their performance. Subsequently, three aerospace test-cases (propulsion performance, structural analysis and re-usable launch vehicle flight simulation) are detailed and the obtained surrogate modeling performance are analyzed. In the fifth section the obtained results are described and discussed. Finally, in the sixth section, the conclusions that may be drawn from what is presented in this chapter are discussed and a few relevant recommendations and future perspectives are provided.

2 Problem Statement

In its most generic formulation, the function f that is being modeled is assumed to depend simultaneously on both continuous and discrete variables, and can be defined as follows:

$$\begin{aligned} y &= f(\mathbf{x}, \mathbf{z}) \\ \text{with } y &\in \mathcal{F}_y \subseteq \mathbb{R} \\ \mathbf{x} &\in \mathcal{F}_x \subseteq \mathbb{R}^q \\ \mathbf{z} &\in \mathcal{F}_z \end{aligned} \tag{1}$$

where $\mathbf{x} = (x_1, \dots, x_q)$ is the vector containing the continuous inputs and $\mathbf{z} = (z_1, \dots, z_r)$ is the vector containing the discrete inputs. q and r being the size of the continuous and discrete dimensions of the function f . For the sake of simplicity, the input vector containing both continuous and discrete variables is represented with the following notation: $\mathbf{w} = \{\mathbf{x}, \mathbf{z}\}$. Each discrete variable z_j is characterized by b_j possible values, also known as levels, which therefore results in a total number of categorical combinations, or categories, of $m = \prod_{k=1}^{k=r} b_k$. In general, two types of discrete variables can be distinguished: ordered and unordered ones [1]. As the name suggests, ordered discrete variables are characterized by a relation of order between their levels. Common examples of ordered variables are ‘small/medium/big’ kind of variables and integer numbers (e.g., number of engines). Unordered discrete variables, instead, are characterized by a lack of relation between their values (e.g., types

of materials, colors). In case only ordered discrete variables are considered, the concept of distance between two points can be adapted and applied. For instance, one can assume that a launcher characterized by 4 boosters may have a performance more similar to a launcher with 3 boosters than to one with only 2, as the ‘distance’ between solutions characterized by 2 and 4 boosters is larger than the one between architectures characterized by 3 and 4 boosters. However, this is not true for unordered variables, because in this case no relation of order exists. For instance, considering the choice of material for a given sub-system, it is not possible to determine *a-priori* the difference between the responses for solutions characterized by aluminum alloys and composite materials.

The core concept of surrogate modeling is to predict the response value y^* of a black-box kind of function f for a generic unmapped input $\mathbf{w}^*$ through an inductive procedure. The surrogate model is created by processing a so-called training data set $\mathcal{D}$, or Design of Experiments (DoE), of n samples $\{\mathbf{x}^i, \mathbf{z}^i, y^i\}$ with $i \in \{1, \dots, n\}$. $\mathcal{D}$ can be defined as follows:

$$\mathcal{D} = \{\mathcal{X} = \{\mathbf{x}^1, \dots, \mathbf{x}^n\}, \quad \mathcal{Z} = \{\mathbf{z}^1, \dots, \mathbf{z}^n\}, \quad \mathcal{Y} = \{y^1, \dots, y^n\}\}$$

where $\mathcal{X}$ and $\mathcal{Z}$ are the subsets of $\mathcal{D}$ containing respectively the continuous and the discrete variables, while $\mathcal{Y}$ is the subset of $\mathcal{D}$ containing the corresponding responses. Generally speaking, the larger the training data set is, the more accurate the model will be. However, it is also important to note that the choice of samples that are included in the training set influences the modeling accuracy. It is therefore common practice to assess the modeling performance over several DoE generated through stochastic processes, such as Latin Hypercube Sampling (LHS), so that the influence of the samples choice is reduced to a minimum.

Most Gaussian process-based modeling techniques have been developed for functions which depend solely on continuous variables, and are therefore not adapted to deal with the presence of discrete variables. A number of different alternatives exist in order to be able to perform Gaussian process-based modeling on mixed-variable functions. A commonly adopted technique allowing to include the discrete variables in the modeling process consists in ‘coding’ the discrete variables in such a way that they can be treated as continuous or numerical variables within the Gaussian process. Alternatively, the solution for the surrogate modeling of mixed continuous/discrete functions that is discussed in this chapter is the possible adaptation of existing continuous Gaussian process surrogate modeling techniques for mixed-variable functions, by modifying their working principle. A few different alternative adaptations have been proposed in the literature and are discussed and compared in the following sections of the chapter.

3 Gaussian Process-Based Surrogate Modeling

As previously mentioned, the idea behind surrogate modeling is to be able to predict the response value y^* of a black-box kind of function for a generic unmapped

input $\mathbf{x}^*$ ($\mathbf{w}^*$ in the mixed variable case) through an inductive procedure, by processing a set of samples $\mathcal{D}$. A family of techniques commonly used for the creation of surrogate models are the Gaussian processes. In this section, the generic concept of Gaussian process-based surrogate modeling for functions depending solely on continuous variables is described. Subsequently, the existing adaptations of Gaussian processes for the surrogate modeling of mixed continuous/discrete function are discussed and compared from a theoretical perspective.

3.1 Continuous Gaussian Process-Based Surrogate Modeling

In its most generic definition, a *Gaussian process* is a generalization of the Gaussian probability distribution [32]. Instead of describing the probability distribution of random scalar or vectorial variables, a Gaussian process characterizes the stochastic properties of a function by mapping the Cartesian product of the probability space and an index space onto an output space. In other words, a generic Gaussian process maps the probability distribution of the possible regression functions. A realization of the process can be defined as:

$$Y(\mathbf{x}, \omega) : \begin{cases} \mathcal{F}_x \times \Omega \rightarrow \mathcal{F}_y \\ (\mathbf{x}, \omega) \rightarrow y(\mathbf{x}, \omega) \end{cases} \quad (2)$$

where $\mathcal{F}_x$ and $\mathcal{F}_y$ are the input and output spaces, Ω is the sample space characterized by a σ -algebra and a probability measure P , y is a realization of Y and finally ω is an event of Ω . The Gaussian process is defined by its mean function μ :

$$\mu(\mathbf{x}) = E[Y(\mathbf{x}, \omega)], \quad \forall \mathbf{x} \in \mathcal{F}_x \quad (3)$$

and its covariance function:

$$Cov(\mathbf{x}^i, \mathbf{x}^j) = E[(Y(\mathbf{x}^i, \omega) - \mu(\mathbf{x}^i))(Y(\mathbf{x}^j, \omega) - \mu(\mathbf{x}^j))], \quad \forall \{\mathbf{x}^i, \mathbf{x}^j\} \in \mathcal{F}_x \times \mathcal{F}_x \quad (4)$$

In practice, a Gaussian process can be defined as [35]:

$$Y(\mathbf{x}, \omega) = \beta(\mathbf{x}) + \varepsilon(\mathbf{x}, \omega) \quad (5)$$

where $\beta(\mathbf{x})$ is a user-defined regression function while $\varepsilon(\mathbf{x}, \omega)$ represents the stochastic estimate of the modeling error with a null mean and an unknown covariance function. In case no information regarding the modeled function f is known, a constant regression function β is usually used [36]:

$$Y(\mathbf{x}, \omega) = \beta + \varepsilon(\mathbf{x}, \omega) \quad (6)$$

A main characteristic of Gaussian processes is that the modeling error is defined in such a way to depend on the covariance between points on the input space. In this

chapter, the Gaussian processes are considered to be invariant by translation (i.e., stationary). As a consequence, the covariance can be re-written as a function of the spatial correlation [32]:

$$\text{Cov}(\mathbf{x}^i, \mathbf{x}^j) = \sigma^2 R(|\mathbf{x}^i - \mathbf{x}^j|) \quad (7)$$

where R is a generic correlation function depending on the Euclidean distance between samples in the search space and σ is the variance of the Gaussian process. A consequence of the stationarity assumption is that the modeling error that is obtained when modeling the response for two points which are close to each other in the search space is similar, while the error resulting from the modeling of the response for two points which are far away from each other is potentially different. Stationary Gaussian processes are often referred to as ‘Kriging’ [29, 34]. The definition of the correlation function to be used in the creation of the surrogate model can be defined by the user. Depending on the characteristics of the function that is being modeled, some choices result more suitable than others. Two popular correlation functions are the *p-exponential* [32] and the *Matern* [27], defined below:

$$\begin{cases} R(\mathbf{x}^i, \mathbf{x}^j)_{p\text{-exp}} = \exp\left(-\sum_{k=1}^{k=q} \theta_k |x_k^i - x_k^j|^{p_k}\right) \\ R(\mathbf{x}^i, \mathbf{x}^j)_{\text{Matern}} = \prod_{k=1}^{k=q} \frac{1}{2^{\nu-1}\Gamma(\nu)} (2\sqrt{\nu}\theta_k |x_k^i - x_k^j|)^{\nu} H_{\nu}(2\sqrt{\nu}\theta_k |x_k^i - x_k^j|) \end{cases} \quad (8)$$

where x_k^i is the value of the variable corresponding to the k th dimension of the data point $\mathbf{x}^i$. θ_k , p_k (or θ_k and ν if the *Matern* function is considered) are hyperparameters characterizing the surrogate model while H is a modified Bessel function. In the remainder of the chapter, the *p-exponential* correlation function is used. The previously defined hyperparameters can be determined by maximizing the so-called likelihood function, which represents the probability for the surrogate model to predict the given data. The likelihood function is defined as follows:

$$L(\boldsymbol{\theta}, \mathbf{p}) = \frac{1}{(2\pi\sigma^2)^{n/2}\sqrt{\det(\mathbf{R})}} \exp\left(-\frac{(\mathbf{y} - \boldsymbol{\beta})^T \mathbf{R}^{-1}(\mathbf{y} - \boldsymbol{\beta})}{2\sigma^2}\right) \quad (9)$$

with:

$$\begin{cases} \boldsymbol{\beta} = \frac{\mathbf{1}^T \mathbf{R}^{-1} \mathbf{y}}{\mathbf{1}^T \mathbf{R}^{-1} \mathbf{1}} \\ \sigma^2 = \frac{(\mathbf{y} - \mathbf{1}\boldsymbol{\beta})^T \mathbf{R}^{-1}(\mathbf{y} - \mathbf{1}\boldsymbol{\beta})}{n} \end{cases} \quad (10)$$

where $\mathbf{R}$ is the $n \times n$ correlation matrix containing the values of the correlation between every point of the training data set:

$$\mathbf{R}_{i,j} = R(\mathbf{x}^i, \mathbf{x}^j) \quad (11)$$

$\mathbf{y}$ is a $n \times 1$ vector containing the responses corresponding to the n training samples, $\mathbf{1}$ is a $n \times 1$ vector of ones and finally $\boldsymbol{\theta}$ and $\mathbf{p}$ are the vectors containing the hyper-

parameters of the surrogate model. For the sake of clarity, in the remainder of this chapter the generic set of hyperparameters characterizing a given surrogate model is referred to with the symbol θ (while $\mathbf{p}$ and other hyperparameters are omitted).

Having determined the hyperparameters that result in an optimal modeling of the data, the surrogate model estimate of the function at an unmapped location $\mathbf{x}^*$ is computed as the mean Gaussian process:

$$y^* = \hat{Y}(\mathbf{x}^*) = \beta + \psi_{\mathcal{X}}^T(\mathbf{x}^*)\mathbf{R}^{-1}(\mathbf{y} - \beta) \quad (12)$$

where $\psi_{\mathcal{X}}$ is an $n \times 1$ vector containing the correlation values between each sample of the training data set $\mathcal{X}$ and the point at which the function is estimated:

$$\psi_{\mathcal{X}_i}(\mathbf{x}^*) = R(\mathbf{x}^*, \mathbf{x}^i) \quad \text{for } i = 1, \dots, n \quad (13)$$

Furthermore, the estimate of the modeling uncertainty associated with a given prediction $\hat{Y}(\mathbf{x}^*)$ can be computed under the form of a mean squared error as:

$$s^2(\mathbf{x}^*) = \sigma^2 (1 - \psi_{\mathcal{X}}^T(\mathbf{x}^*)\mathbf{R}^{-1}\psi_{\mathcal{X}}(\mathbf{x}^*)) \quad (14)$$

Up to this point, only Gaussian processes for continuous variables were discussed. In case the modeled function depends simultaneously on continuous and discrete variables, these surrogate modeling techniques cannot be directly applied. A first approach, commonly used in the engineering domain, consists in creating a separate surrogate model for every category of the considered function. Every surrogate model is therefore trained with the samples of the DoE which belong to the modeled category. Alternatively, if an actual mixed continuous/discrete surrogate model of the function is required, an adaptation of the surrogate model definition is necessary. As mentioned in the previous section, generic discrete values might be characterized by an unordered nature and by consequence it is not possible to directly apply the concept of Euclidean distance when computing the correlation value between two samples. Three different approaches for the definition of a mixed continuous/discrete correlation function $R^*(\mathbf{w}^i, \mathbf{w}^j)$ exist. The first one consists in transforming the discrete variables characterizing the problems into continuous ones through a ‘coding’ process and subsequently use one of the previously defined correlation functions for continuous variables. The second approach consists in re-defining the concept of distance between two points so that it can be applied in the mixed continuous/discrete search space. Finally, the last approach consists in defining the correlation as a combination of two separate terms, respectively characterizing the influence of the continuous and discrete variables of the two considered data samples. In the following sections, the above-mentioned alternatives for the surrogate modeling of mixed continuous/discrete functions are described, compared and commented.

3.2 *Category-Wise Gaussian Process Based Surrogate Modeling*

The most commonly used approach to avoid dealing with the surrogate modeling of a mixed continuous/discrete function consists in creating a separate surrogate model (Gaussian process in this case) for every category of the modeled function, thus splitting the mixed variable surrogate modeling problem into m independent continuous surrogate modeling problems. Every Gaussian process is trained separately by using the data samples which belong to the modeled category. However, this approach becomes quickly unfeasible for problems characterized by a large number of discrete categories, as a considerable amount of data becomes necessary in order to provide an accurate enough surrogate modeling of every discrete category. The computation time required for the training of the m Gaussian processes may also be an issue. Furthermore, a large amount of information regarding possible existing correlations between the various categories and discrete levels remains unused. Additionally, this approach is computationally inefficient due to the fact that identical or similar operations are performed for every separate model that is created. When this approach is considered, the total number of hyperparameters that must be determined during the creation of the surrogate models is equal to $m \cdot 2 \cdot q$ and increases therefore linearly with the number of categories which characterize the modeled function. For the remainder of the chapter, the category-wise Gaussian process based mixed continuous/discrete surrogate modeling is used as a reference in order to properly assess the advantages of creating mixed variable Gaussian processes rather than a large number of continuous ones.

3.3 *Coding Based Gaussian Process Based Surrogate Modeling*

In order to avoid creating a Gaussian process surrogate model for every category of the considered problem, a commonly adopted option is to transform the discrete variables into continuous ones through a coding process. A number of different codings for discrete variables exists in the literature [18], each one with its advantages, limitations and constraints. In the following paragraph, two of the more commonly used codings, the direct conversion and the dummy coding, are described and discussed.

Table 1 Direct conversion of the discrete variable characterizing the material choice

Material	Direct conversion variable value
Steel	0
Aluminum alloy	1
Titanium alloy	2
Composite material	3

Table 2 Dummy coding of the discrete variable characterizing the material choice

Material	Dummy variables values			
Steel	0	0	0	1
Aluminum alloy	0	0	1	0
Titanium alloy	0	1	0	0
Composite material	1	0	0	0

The direct conversion [20] is the most intuitive coding for discrete variables. Its principle consists in assigning a numerical value to every level of the discrete variable. For instance, considering the choice of the material to be used for a given system, with the possible options being steel, aluminum alloy, titanium alloy and composite material, a possible direct conversion of the discrete variable is shown in Table 1. However, this numerical conversion implicitly introduces a relation of order between the variable levels which may have no physical meaning. For instance, in the previously presented numerical conversion, the ‘distance’ between two solutions characterized by aluminum alloy and titanium alloy, respectively, is smaller than the distance between two solutions characterized by aluminum alloy and composite material. This difference has no physical meaning and therefore introduces a bias in the surrogate modeling. This aspect can be further accentuated by the fact that the numerical values which are assigned to the discrete variable levels are user-defined. A slightly more refined type of coding that can be encountered in the literature is the so-called dummy coding [37]. Its functioning principle is to represent a discrete variable characterized by b_k levels with a binary vector of b_k ‘dummy variables’. All the values of this vector are equal to zero, except for one, corresponding to the selected level among all the possible choices, which is equal to one. The dummy coding of the previously described discrete variable characterizing the choice of material for a given subsystem is shown in Table 2.

This solution results in having all the different options for a given discrete variable at the same distance from the others and therefore allows to include both ordered and unordered discrete variables in the modeling process without user-induced bias.

Having transformed all the discrete variables characterizing the problem into continuous one through a coding process, it is possible to perform the surrogate modeling of the considered function by using a standard continuous Gaussian process. For instance, if the direct conversion coding is used, the *p-exponential* correlation function for the mixed continuous/discrete variables can be re-written as:

$$R^*(\mathbf{w}^i, \mathbf{w}^j)_{p\text{-exp}} = \exp \left(- \sum_{k=1}^{k=q} \theta_k |x_k^i - x_k^j|^{p_k} - \sum_{k=1}^{k=r} \theta_{k+q} |\tilde{z}_k^i - \tilde{z}_k^j|^{p_{k+q}} \right) \quad (15)$$

where a generic $\tilde{z}_k^i$ represents the continuous variable resulting from the coding of the discrete variable $\tilde{z}_k^i$. The number of hyperparameters θ and p that characterize the

resulting correlation depends on the type of coding that is used. The remainder of the creation of the Gaussian process for mixed continuous/discrete functions is identical to the one described by Eqs. (9)–(13). The use of discrete variable coding represents a simple alternative for the surrogate modeling of mixed continuous/discrete problems and allows the user to rely on standard continuous variable Gaussian processes. However, for complex functions this solution results too simplistic and does not provide the required surrogate modeling accuracy. Furthermore, if coding techniques such as the dummy coding are used, the number of additional variables which are created process is proportional to the number of levels that characterize the modeled function discrete variables, which can become problematic for complex problems and large sets of data. In order to accurately model complex mixed variable problems, proper adaptations of the Gaussian processes to the presence of discrete variables are therefore required.

3.4 Correlation Function Based on Alternative Definitions of Distance in the Mixed Continuous/Discrete Search Space

A Gaussian process relying on an alternative definition of the distance in the mixed continuous/discrete search space is proposed by M. Halstrup [16]. The distance definition used in this case is the so-called ‘Gower distance’, first proposed by J. Gower [13], in which the coordinates of the two points that are being considered are compared dimension-wise. For the continuous dimensions, the distance is proportional to the Manhattan distance, while for the discrete variables the distance is a weighted binary value, depending on the similarity between the variable values. In practice, Gower distance between two points can be expressed as follows:

$$d_{gow}(\mathbf{w}^i, \mathbf{w}^j) = \frac{\sum_{k=1}^{k=q} \frac{|x_k^i - x_k^j|}{\Delta x_k}}{r + q} + \frac{\sum_{k=1}^{k=r} s(z_k^i, z_k^j)}{r + q} \quad (16)$$

where Δx_k is the range of the continuous variable in the k -dimension and s is a score function defined as:

$$s(z_k^i, z_k^j) = \begin{cases} 0 & \text{if } z_k^i = z_k^j \\ 1 & \text{if } z_k^i \neq z_k^j \end{cases} \quad (17)$$

Having re-defined the distance in order to take into account the presence of both continuous and discrete variables, the correlation function can be adapted to mixed-variable problems. Considering, for instance, the p -exponential correlation function, it may be re-written as:

$$R^*(\mathbf{w}^i, \mathbf{w}^j)_{p-exp} = \exp \left[- \sum_{k=1}^{k=q} \theta_k \left(\frac{|x_k^i - x_k^j|}{\Delta x_k} \right)^{p_k} - \sum_{k=1}^{k=r} \theta_{k+q} \left(\frac{s(z_k^i, z_k^j)}{r + q} \right)^{p_{k+q}} \right] \quad (18)$$

The remainder of the creation of the Gaussian process based surrogate model for mixed continuous/discrete functions is identical to the one described by Eqs. (9)–(13) and requires no further adaptation.

The Gower distance based Gaussian process is characterized by $2(q + r)$ hyperparameters that must be determined for the characterization of the surrogate model. The adaptation of a standard Gaussian process into a Gower distance based one is relatively simple. However, it must also be noticed that because each discrete variable is only parameterized by 2 hyperparameters θ, p , the surrogate model may present poor modeling performances in the presence of discrete variables with large numbers of discrete levels. For the same reason, the simultaneous presence of correlation and anti-correlation trends described by the same discrete variable are also difficult to model for the Gower distance based Gaussian process.

3.5 Combined Continuous/Discrete Correlation Functions

An alternative approach for the definition of a correlation function between data points in the mixed continuous/discrete search space consists in combining two separate terms, each one representing the influence of either the continuous or the discrete variables of the two considered data samples. Such a correlation function is proposed by Q. Zhou et al. [41], as the product between the *p-exponential* correlation function computed on the continuous variables and a parameter characterizing the correlation between the discrete categories of the two data samples:

$$R^*(\mathbf{w}^i, \mathbf{w}^j) = R(\mathbf{x}^i, \mathbf{x}^j) \mathcal{T}_{c_i, c_j} \quad (19)$$

where $\mathcal{T}_{c_i, c_j}$ is the coefficient characterizing the correlation between the two discrete categories c_i and c_j . For the sake of simplicity, the parameters $\mathcal{T}$ are grouped in an $m \times m$ matrix $\mathbf{T}$, presenting ones on the diagonal and $\mathcal{T}_{c_i, c_j}$ on the element $\{i, j\}$. In order to ensure that the correlation matrix $\mathbf{R}^*$ is valid and invertible, $\mathbf{T}$ must be a positive definite matrix with unit diagonal (PDUDE) [30]. This characteristic can be ensured by constructing the matrix through a so-called ‘hypersphere decomposition’ [33]. More specifically, $\mathbf{T}$ is defined through a Cholesky decomposition as:

$$\mathbf{T} = \mathbf{L}\mathbf{L}^T \quad (20)$$

where, $\mathbf{L}$ is a lower triangular matrix in which the elements of the k th row represent the coordinates of a point on the surface of a k -dimensional sphere with a unit radius. The triangular matrix elements $l_{k,s}$ are defined as follows:

$$\begin{cases} l_{1,1} = 1 \\ l_{k,1} = \cos(\alpha_{k,1}) \\ l_{k,s} = \sin(\alpha_{k,1}) \dots \sin(\alpha_{k,s-1}) \cos(\alpha_{k,s}) & \text{for } s = 2, \dots, k-1 \\ l_{k,k} = \sin(\alpha_{k,1}) \dots \sin(\alpha_{k,k-2}) \sin(\alpha_{k,k-1}) \end{cases} \quad (21)$$

where $\alpha_{k,s} \in (0, \pi)$ are the hyperparameters characterizing the correlation between the various discrete categories. Although there is a total of m^2 combinations of discrete categories, thanks to the inherent symmetrical nature of the correlation and to unitary correlation values between identical categories, $m(m-1)/2$ parameters $\alpha_{k,s}$ are required in order to define the matrix $\mathbf{L}$, resulting in a total of $2q + m(m-1)/2$ hyperparameters characterizing the surrogate model.

As an illustrative example, considering a function characterized by 2 discrete variables with 2 levels each ($z_1 \in \{0, 1\}$, $z_2 \in \{0, 1\}$) resulting in 4 categories, the lower triangular Cholesky decomposition matrix $\mathbf{L}$ depends on 6 hyperparameters α and has the following expression:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ \cos(\alpha_{2,1}) & \sin(\alpha_{2,1}) & 0 & 0 \\ \cos(\alpha_{3,1}) & \sin(\alpha_{3,1}) \cos(\alpha_{3,2}) & \sin(\alpha_{3,1}) \sin(\alpha_{3,2}) & 0 \\ \cos(\alpha_{4,1}) & \sin(\alpha_{4,1}) \cos(\alpha_{4,2}) & \sin(\alpha_{4,1}) \sin(\alpha_{4,2}) \cos(\alpha_{4,3}) & \sin(\alpha_{4,1}) \sin(\alpha_{4,2}) \sin(\alpha_{4,3}) \end{bmatrix} \quad (22)$$

When determining the Gaussian process hyperparameters, the hypersphere decomposition enables to ensure that the matrix $\mathbf{R}^*$ is invertible all along the likelihood function optimization process. Compared to the Gower distance based Gaussian process, this method can theoretically provide a more accurate surrogate modeling of mixed continuous/discrete functions, as it characterizes a separate correlation parameter for every discrete category combination of the problem. However, when one or more categories are not present in the training set, it is not possible to compute the matrix $\mathbf{L}$ as not all the necessary data is available for the surrogate model to determine the optimal values for the set of hyperparameters θ . Furthermore, for problems characterized by a large combination of discrete categories, the number of hyperparameters that must be determined becomes very large and by consequence considerably hard to handle in the maximum likelihood optimization problem. In order to avoid these problems, Q. Zhou et al. also proposed a ‘*simplified*’ dimension-wise variant of the hypersphere decomposition based correlation function [41]. In this case, the correlation in the discrete domain is processed separately for every dimension and one correlation parameter is therefore computed for every combination of levels of the same discrete variables. This results in the following expression of the correlation:

$$R^*(\mathbf{w}^i, \mathbf{w}^j) = R(\mathbf{x}_i, \mathbf{x}_j) \prod_{k=1}^{k=r} \mathcal{T}_{z_k^i, z_k^j} \quad (23)$$

where $\mathcal{T}_{z_k^i, z_k^j}$ represents the correlation parameter between the values z_k^i and z_k^j of the k th discrete variable. Each one of the r matrices $\mathbf{T}_k$ characterizing the correlations between the various discrete levels of the k th discrete variable is a $b_k \times b_k$ PDUDE matrix constructed by using the same hypersphere decomposition shown in Eqs. (20)–(21). When using this dimension-wise version of the hypersphere decomposition

based correlation function, the number of hyperparameters necessary in order to define the set of r matrices $\mathbf{T}_k$ is equal to $\sum_{k=1}^{k=r} b_k(b_k - 1)/2$, resulting in a total of $2q + \sum_{k=1}^{k=r} b_k(b_k - 1)/2$ hyperparameters characterizing the surrogate model. As an illustrative example, for a function characterized by 2 discrete variables with 2 levels each ($z_1 \in \{0, 1\}$, $z_2 \in \{0, 1\}$) resulting in 4 categories, the lower triangular Cholesky decomposition matrices used for the computation of $\mathbf{T}_1$ and $\mathbf{T}_2$, depend on 1 hyperparameter α each, for a total of 2 hyperparameters. The two matrices $\mathbf{L}_1$ and $\mathbf{L}_2$ have the following expression:

$$\mathbf{L}_1 = \left[\begin{array}{c|c} 1 & 0 \\ \hline \cos(\alpha_{2,1}) & \sin(\alpha_{2,1}) \end{array} \right] \quad \mathbf{L}_2 = \left[\begin{array}{c|c} 1 & 0 \\ \hline \cos(\alpha_{2,1}^*) & \sin(\alpha_{2,1}^*) \end{array} \right] \quad (24)$$

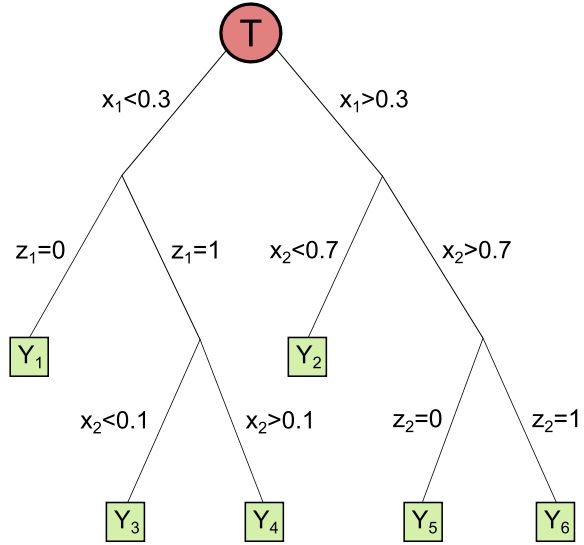
When compared to the standard hypersphere decomposition, the dimension-wise variant offers a better scaling with the discrete dimension of the problem that is being modeled in terms of number of hyperparameters to be determined, but as a trade-off provides a theoretically less accurate modeling of the correlation between the various discrete categories of the problem. Thanks to the fact that the discrete correlation parameters are determined separately for every dimension rather than for every discrete category, this variant of the hypersphere decomposition Gaussian process does not require all the discrete categories of the modeled function to be present in the training data set in order to create the surrogate model. The remainder of the creation of the Gaussian process based surrogate model for mixed continuous/discrete functions is identical to the one described by the Eqs. (9)–(13) and requires no further adaptation.

In the previous paragraphs, 2 opposite approaches to the surrogate modeling of mixed continuous/discrete functions were presented. The first one consisted in creating a separate surrogate model for every discrete category of the problem, while the second consisted in creating a single model characterizing the entirety of the responses of the considered function. A middle-ground between these 2 approaches would be to create a limited number of separate surrogate models by regrouping the discrete variables which can be explained by the same Gaussian process while separating the ones presenting behaviors which are too uncorrelated to be modeled together. In the following section, the Bayesian Treed Gaussian Process, a surrogate modeling method relying on this kind of approach, is presented and discussed.

3.6 Dummy Coding Based Bayesian Treed Gaussian Process

The dummy coding Bayesian Treed Gaussian Process (BTGP) relies on a tree-like binary partitioning of the modeled function search space into a number of complementary subsets [14]. In each one of these subsets, a separate Kriging model is defined. In case the search space subset that is considered is defined in the mixed variable domain, dummy coding of the discrete variables is used. This approach allows to

Fig. 1 Example of a partition tree for a generic function characterized by 2 continuous variables and 2 discrete variables having 2 levels each. Y_i are the local Gaussian processes



properly describe the trend of functions which present different behaviors according to the regions of the search space that are being considered [23]. However, it also makes the surrogate model creation more complex, as it requires to determine the most coherent tree-like partition of the search space. Please note that, to the best of the authors' knowledge, no other adaptation of the BTGP for mixed continuous/discrete variables has been developed, excepted for the dummy coding based one present in the R 'mlr' package [5] and considered in this chapter. For illustrative purposes, an example of mixed variable partition tree for a generic function characterized by 2 continuous variables and 2 discrete variables with 2 levels each is shown in Fig. 1. The local Gaussian processes Y_i are defined in a subset of the modeled function domain determined by the series of 'binary splits' characterizing the terminal node they belong to.

The optimal partition tree is estimated by maximizing its probability to describe the modeled function response with respect to the available data set. In practice, the partition tree and the local Kriging models are optimized simultaneously as the arguments of the following function:

$$p(\mathcal{Y}|\mathcal{X}, \mathcal{Z}, T) = \int p(\mathcal{Y}|\mathcal{X}, \mathcal{Z}, \boldsymbol{\theta}, T)p(\boldsymbol{\theta}|T)d\boldsymbol{\theta} \quad (25)$$

where T is the partition tree and $\boldsymbol{\theta}$ refers to the set of hyperparameters characterizing the entirety of the Kriging surrogate models defined in the search space subsets. In the original implementation of the BTGP, a Markov Chain Monte Carlo (MCMC) is used to create the sequence of partitions trees T which converges to the posterior of Eq. 25:

$$p(T|\mathcal{X}, \mathcal{Z}, \mathcal{Y}) \quad (26)$$

A population of partitions trees is initialized to null trees T_0 . Then, at every iteration i of the optimization, a candidate tree T^* is created with a probability $q(T^i, T^*)$ by modifying the incumbent solution with an operation randomly chosen among the following ones: growing, pruning, changing and branch swapping, which are defined as follows [16]:

- Growing: splitting a terminal node into 2 new ones according to a user-defined splitting rule.
- Pruning: transforming a parent node into a terminal node by dropping the attached branches.
- Changing: re-defining the splitting rule of a randomly chosen internal node
- Swapping: Interchanging the splitting rule of two randomly chosen non-terminal nodes.

Having characterized T^* , the probability for it to replace the incumbent solution is equal to:

$$p^*(T^i, T^*) = \min \left\{ \frac{q(T^*, T^i)p(\mathcal{Y}|\mathcal{X}, \mathcal{Z}, T^*)p(T^*)}{q(T^i, T^*)p(\mathcal{Y}|\mathcal{X}, \mathcal{Z}, T^i)p(T^i)}, 1 \right\} \quad (27)$$

otherwise, the incumbent solution remains unchanged: $T^{i+1} = (T^i)$. It is possible to include a constraint specifying a minimum number of data samples per terminal node of the tree in order to ensure a baseline surrogate modeling accuracy over the entirety of the search space. At the end of the previously described MCMC optimization, both the partition tree structure and the characterization of a local kriging for every ‘branch’ of said tree is determined. The BTGP prediction works very similarly to the standard Kriging prediction process described in Eqs. (9)–(13), with the only difference being that the correct local model corresponding to the search space area that is being considered needs to be used. Compared to the previously discussed mixed continuous/discrete Gaussian processes, the BTGP allows to accurately model functions which present considerable variations in their behavior over the span of the search space by relying on an optimal partition of the domain. Furthermore, the creation of the single local surrogate models results less complex when compared to the comprehensive surrogate models, as a smaller number of hyperparameters must be determined and smaller data sets must be processed. However, to the best of the authors’ knowledge, the only existing adaptation of the BTGP to mixed continuous/discrete surrogate modeling relies on the dummy coding of the discrete variables, which, as previously discussed, might result too simplistic when dealing with complex problem.

3.7 Surrogate Model Comparison

In order to provide the reader with a better overview of the characteristics of the various mixed continuous/discrete surrogate modeling techniques presented in the

Table 3 Comparison of surrogate modeling techniques. The method numbers refer to: category-wise Gaussian process [1], dummy coding based Gaussian process [2], Gower distance Gaussian process [3], hypersphere decomposition Gaussian process [4], dimension-wise hypersphere decomposition Gaussian process [5], BTGP [6]. The number of hyperparameters is calculated by considering a *p-exponential* correlation function

Method	Hyperparameters	Strengths	Weaknesses
1	$2 \cdot q \cdot m$	No adaptation required	Poor modeling Large data sets required
2	$2q + \sum_{k=1}^{k=r} 2b_k$	Simple implementation	Poor modeling of discrete variables
		No adaptation required	Additional variables to process
3	$2(q + r)$	Simple implementation	Poor modeling of category-wise correlations
		Low N° of hyperparameters	
4	$2q + m(m - 1)/2$	Accurate modeling of category-wise correlations	Large N° of hyperparameters
			Poor scaling with N° of categories
			All categories must be present in the DoE
5	$2q + \sum_{k=1}^{k=r} b_k(b_k - 1)/2$	Good scaling with N° of categories	No category-wise correlation parameter
6	Varying	Good scaling with problem size	Challenging partition-tree creation
		Accurate modeling of local trends	Overly simplistic dummy coding
		Simple local surrogate models	Complex

previous sections, a summary of their relative strengths and weaknesses is presented in Table 3.

It can be noted from the summary table that the number of hyperparameters characterizing a given surrogate model does not vary in the same way with respect to the number of discrete variables and with respect to the number of discrete levels that characterize them. For illustrative purposes, the number of hyperparameters describing the surrogate model of an hypothetical function characterized by 2 continuous variables and a varying discrete dimension is provided in Fig. 2. Two distinct separate cases are considered. In the first one, the function depends on a varying number of discrete variables, each one characterized by 3 discrete levels. In the second case, instead, the considered function depends on 3 discrete variables characterized by

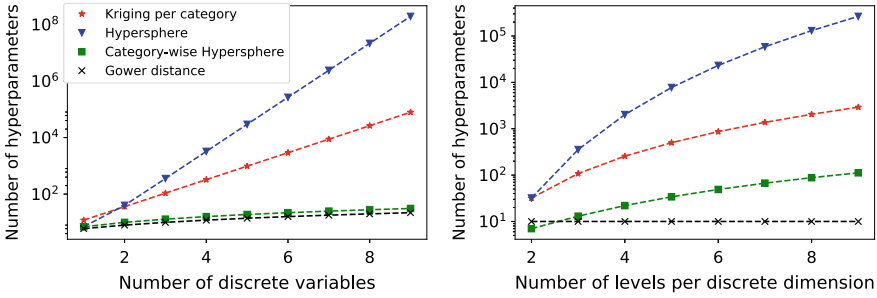


Fig. 2 Number of hyperparameters required for the surrogate modeling of a function characterized by 2 continuous variables and a varying discrete dimension considering a p -exponential correlation function. On the left, the function depends on a varying number of discrete variables, each one characterized by 3 levels. On the right, the function depends on 3 discrete variables that are characterized by a varying number of discrete levels

a varying number levels (the same for all the variables). As previously mentioned, it can be seen that except for particularly low discrete dimensions, the hypersphere decomposition Gaussian process is the surrogate modeling techniques with the largest number of hyperparameters. The two techniques which seem to scale better with the discrete dimension of the modeled function, in terms of number of hyperparameters, are the Gower distance and the category-wise hypersphere decomposition Gaussian processes.

4 Results

In order to assess the performance of the various surrogate modeling methods as well as their dependence on the dimensions (both continuous and discrete) of the modeled function, six different test-cases of varying complexity and dimension are used. Three test-cases are analytical benchmark functions, adapted in order to present a dependency on both continuous and discrete variables, while the other three test-cases are ‘real world’ aerospace design simulations. More specifically, the considered aerospace test-cases are an analysis of a launcher engine propulsion performance, a structural analysis of a launcher thrust frame and a flight simulation of a re-usable launch vehicle system. Every benchmark is tested for three different data set sizes, which depend on the dimension of the specific benchmark function and on its number of discrete categories. Furthermore, in order to take into account the dependency of the modeling performances on the training data set, every modeling routine is repeated 10 times on different DoE of the same size, generated through a semi-stochastic LHS and tested on a constant step-size grid on the entire search space. Due to the considerably large computation time required to generate the training data sets for the structural analysis and launch vehicle flight simulations (ranging from several minutes to nearly an hour for each function evaluation), the modeling of these

test-cases is only performed on two separate DoE of the same size (with the testing being performed on the data set which is not used for the training).

Please note that in order to provide a relevant comparison in terms of surrogate modeling accuracy, only Gaussian processes adapted for mixed continuous/discrete functions are considered in the following sections. Therefore, the results that may be obtained with a dummy coding based Kriging are not presented. Furthermore, the category-wise Kriging is considered as the reference surrogate modeling method, due to the fact that it represents the industry standard for what concerns surrogate modeling of mixed variable functions.

For all the results presented in the following sections, excepted the cases in which the BTGP is considered, the hyperparameters characterizing the surrogate models are optimized by using a Covariance Matrix Adaptation Evolution Strategy (CMA-ES) [17]. The number of iterations assigned to the optimizations varies as a function of the number of hyperparameters that must be determined. The hyperparameters characterizing the BTGP are instead optimized at the same time as the partition tree with the help of a MCMC (Eq. 25).

4.1 Branin Function

The first analytical benchmark function to be considered is a modified version of the Branin function characterized by two continuous variables and two discrete variables, each one with 2 levels, thus resulting in overall 4 discrete categories. This version of the Branin function is defined as follows:

$$h(x_1, x_2) = (((15x_2 - \frac{5}{4\pi^2}(15x_1 - 5)^2 + \frac{5}{\pi}(15x_1 - 5) - 6)^2 + 10(1 - \frac{1}{8\pi})\cos(15x_1 - 5) + 10) - 54.8104)/51.9496 \quad (28)$$

$$f(x_1, x_2, z_1, z_2) = \begin{cases} h(x_1, x_2) & \text{if } z_1 = 0 \text{ and } z_2 = 0 \\ 0.4h(x_1, x_2) + 1.1 & \text{if } z_1 = 0 \text{ and } z_2 = 1 \\ -0.75h(x_1, x_2) + 5.2 & \text{if } z_1 = 1 \text{ and } z_2 = 0 \\ -0.5h(x_1, x_2) - 2.1 & \text{if } z_1 = 1 \text{ and } z_2 = 1 \end{cases} \quad (29)$$

with

$$x_1 \in [0, 1], \quad x_2 \in [0, 1], \quad z_1 \in \{0, 1\}, \quad z_2 \in \{0, 1\}$$

The responses of the Branin function for each discrete category are presented in Fig. 3. It can be noticed that the various discrete categories are characterized by a pair-wise anti-correlation and a considerable relative offset, which results challenging for the Gaussian processes to model.

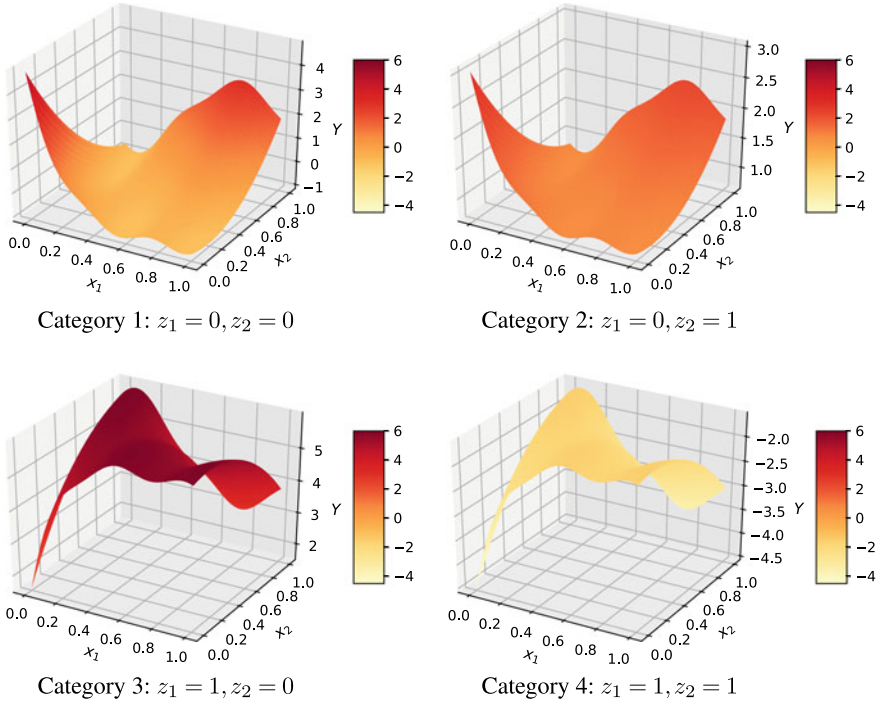


Fig. 3 The 4 discrete categories of the Branin function

This analytically simple benchmark function can be used to make some considerations in order to provide the reader with a better insight on the actual implementation of the modeling methods presented in the previous section. Considering 2 points with the following coordinates:

$$\mathbf{w}_1 = \{0.7, 0.1, 0, 0\}$$

$$\mathbf{w}_2 = \{0.2, 0.5, 1, 0\}$$

which belong to the first and third discrete category, respectively, and assuming 4 generic values of θ_k and p_k , the Gower distance based correlation between the two points is equal to:

$$R^*(\mathbf{w}_1, \mathbf{w}_2) = \exp \left\{ - \left[\theta_1 \left(\frac{|0.7-0.1|}{1} \right)^{p_1} + \theta_2 \left(\frac{|0.1-0.5|}{1} \right)^{p_2} + \theta_3 \left(\frac{1}{4} \right)^{p_3} + \theta_4 \left(\frac{0}{4} \right)^{p_4} \right] \right\} \quad (30)$$

where 1 is the range of variation of the continuous variables and 4 is the global dimension of the function. If an hypersphere decomposition based correlation is considered instead, the correlation value is equal to:

$$R^*(\mathbf{w}_1, \mathbf{w}_2) = \exp \left\{ - \left[\theta_1 (|0.7 - 0.1|)^{p_1} + \theta_2 (|0.1 - 0.5|)^{p_2} \right] \right\} \mathcal{T}_{c_1, c_3} \quad (31)$$

where $\mathcal{T}_{c_1, c_3}$ is the element $\{1, 3\}$ (equal to the element $\{3, 1\}$) of the 4×4 matrix $\mathbf{T}$, which characterizes the correlation between the discrete categories 1 and 3 of the Branin function (see Eq. 22). In this case, the matrix $\mathbf{T}$ depends on $\frac{4 \cdot 3}{2} = 6$ hyperparameters α . Alternatively, if the dimension-wise variant of the hypersphere decomposition based correlation is considered, the following correlation value is obtained:

$$R^*(\mathbf{w}_1, \mathbf{w}_2) = \exp \left\{ - \left[\theta_1 (|0.7 - 0.1|)^{p_1} + \theta_2 (|0.1 - 0.5|)^{p_2} \right] \right\} \mathcal{T}_{1,0} \mathcal{T}_{2,0} \quad (32)$$

In this case, the matrices $\mathbf{T}_1$ and $\mathbf{T}_2$ which contain the elements characterizing the correlation values between the various levels of the discrete variables z_1 and z_2 , respectively, have a size of 2×2 and each one depends on $\frac{2 \cdot 1}{2} = 1$ hyperparameter (see Eq. 24). Please note that according to the definitions provided by Eqs. (20)–(21), $\mathcal{T}_{2,0}$ being an element of the matrix diagonal is equal to 1.

The results obtained when modeling the Branin function with the various mixed continuous/discrete Gaussian processes are provided in Fig. 4. The modeling performance parameter that is used is the Root of the Mean Squared Error (RMSE). The presented box-plots contain the results obtained for 10 different training data sets. The testing is performed with data sets of 40, 80 and 160 samples (i.e., 10, 20 and 40 samples per discrete category). From Fig. 4 it can be noticed that almost all mixed variable surrogate modeling methods result in a better modeling accuracy than the category-wise Kriging. The only exception being the BTGP with a DoE of 40 samples, as in this case the amount of provided data is not sufficient for the training process to determine the optimal partition tree. By consequence, the MCMC routine quickly converges to a sub-optimal partition tree. This can also explain the particu-

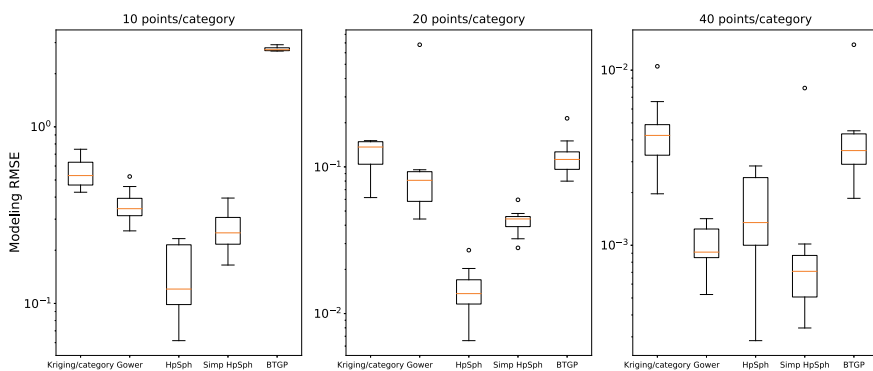


Fig. 4 Modeling RMSE for the Branin function obtained on 3 separate batches of 10 training data sets of varying size. From left to right, the used modeling methods are: a separate Kriging model per discrete category, a Gower distance based Gaussian process, a standard hypersphere decomposition based Gaussian process, a dimension-wise hypersphere decomposition based Gaussian process and a BTGP

larly low standard deviation for this particular result. It can be noticed as well that, in general, the 2 hypersphere decomposition Gaussian processes seem to outperform the other surrogate modeling methods, closely followed by the Gower distance based one.

4.2 Augmented Branin Function

The second analytical benchmark function to be considered is a larger dimension version of the previously described Branin function, characterized by 10 continuous variables and 2 discrete variables, each one with 2 levels, thus resulting in 4 discrete categories. This test-case allows to assess the dependency of the various surrogate models performance on the continuous dimension of the problem that is being considered. The ‘augmented’ Branin function is defined as follows:

$$h(x_i, x_j) = (((15x_j - \frac{5}{4\pi^2}(15x_i - 5)^2 + \frac{5}{\pi}(15x_i - 5) - 6)^2 + 10(1 - \frac{1}{8\pi})\cos(15x_i - 5) + 10) - 54.8104)/51.9496 \quad (33)$$

$$H(x_1, \dots, x_{10}) = h(x_1, x_2) + h(x_3, x_4) + h(x_5, x_6) + h(x_7, x_8) + h(x_9, x_{10}) \quad (34)$$

$$f(x_1, \dots, x_{10}, z_1, z_2) = \begin{cases} H(x_1, \dots, x_{10}) & \text{if } z_1 = 0 \text{ and } z_2 = 0 \\ 0.4H(x_1, \dots, x_{10}) + 1.1 & \text{if } z_1 = 0 \text{ and } z_2 = 1 \\ -0.75H(x_1, \dots, x_{10}) + 5.2 & \text{if } z_1 = 1 \text{ and } z_2 = 0 \\ -0.5H(x_1, \dots, x_{10}) - 2.1 & \text{if } z_1 = 1 \text{ and } z_2 = 1 \end{cases} \quad (35)$$

with

$$x_1, \dots, x_{10} \in [0, 1], \quad z_1 \in \{0, 1\}, \quad z_2 \in \{0, 1\}$$

The results obtained when modeling the augmented Branin function with the various mixed continuous/discrete Gaussian processes are provided in Fig. 5. The testing was performed with data sets of 200, 400 and 800 samples (i.e., 50, 100 and 200 samples per discrete category). It can be noticed that when the continuous dimension of the problem is increased, the mixed variable surrogate modeling methods still tend to perform better than the category-wise Kriging. However, it can be seen that for large enough training data sets, the BTGP and the Gower distance based Gaussian process produce results comparable to the category-wise Kriging. Instead, the 2 hypersphere decomposition Gaussian processes result in a considerably better modeling accuracy when compared to the other methods.

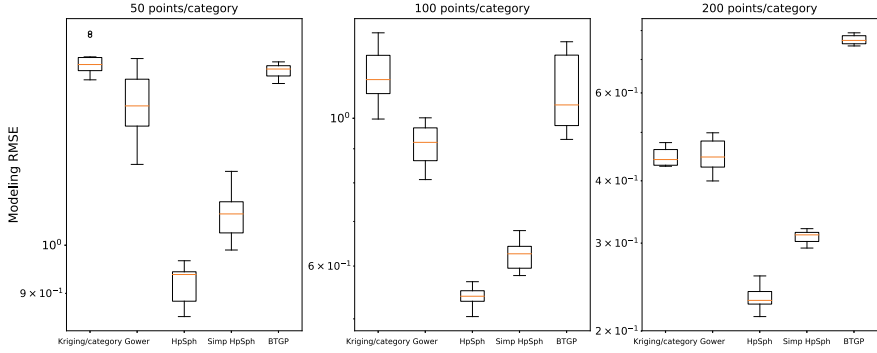


Fig. 5 Modeling RMSE for the augmented Branin function obtained on 3 separate batches of 10 training data sets of varying size. From left to right, the used modeling methods are: a separate Kriging model per discrete category, a Gower distance based Gaussian process, a standard hypersphere decomposition based Gaussian process, a dimension-wise hypersphere decomposition based Gaussian process and a BTGP

4.3 Goldstein Function

The third analytical benchmark function to be considered is a modified version of the Goldstein function characterized by 2 continuous variables and 3 discrete variables, each one with 3 levels, thus resulting in 9 discrete categories. This analytical test-case is introduced in order to assess the dependency of the various surrogate models performance on the number of categories that characterize the problem that is being considered. The Goldstein function is defined as follows:

$$\begin{aligned}
 h(x_1, x_2, x_3, x_4) = & 53.3108 + 0.184901x_1 - 5.02914x_1^3 10^{-6} + 7.72522x_1^4 10^{-8} - \\
 & 0.0870775x_2 - 0.106959x_3 + 7.98772x_3^3 10^{-6} + \\
 & 0.00242482x_4 + 1.32851x_4^3 10^{-6} - 0.00146393x_1x_2 - \\
 & 0.00301588x_1x_3 - 0.00272291x_1x_4 + 0.0017004x_2x_3 + \\
 & 0.0038428x_2x_4 - 0.000198969x_3x_4 + 1.86025x_1x_2x_3 10^{-5} - \\
 & 1.88719x_1x_2x_4 10^{-6} + 2.50923x_1x_3x_4 10^{-5} - \\
 & 5.62199x_2x_3x_4 10^{-5}
 \end{aligned} \tag{36}$$

$$f(x_1, x_2, z_1, z_2) = h(x_1, x_2, x_3, x_4) \tag{37}$$

with

$$x_1 \in [0, 100], \quad x_2 \in [0, 100], \quad z_1 \in \{0, 1, 2\}, \quad z_2 \in \{0, 1, 2\}$$

Table 4 Characterization of the Goldstein function discrete categories

	$z_1 = 0$	$z_1 = 1$	$z_1 = 2$
$z_2 = 0$	$x_3 = 20, x_4 = 20$	$x_3 = 50, x_4 = 20$	$x_3 = 80, x_4 = 20$
$z_2 = 1$	$x_3 = 20, x_4 = 50$	$x_3 = 50, x_4 = 50$	$x_3 = 80, x_4 = 50$
$z_2 = 2$	$x_3 = 20, x_4 = 80$	$x_3 = 50, x_4 = 80$	$x_3 = 80, x_4 = 80$

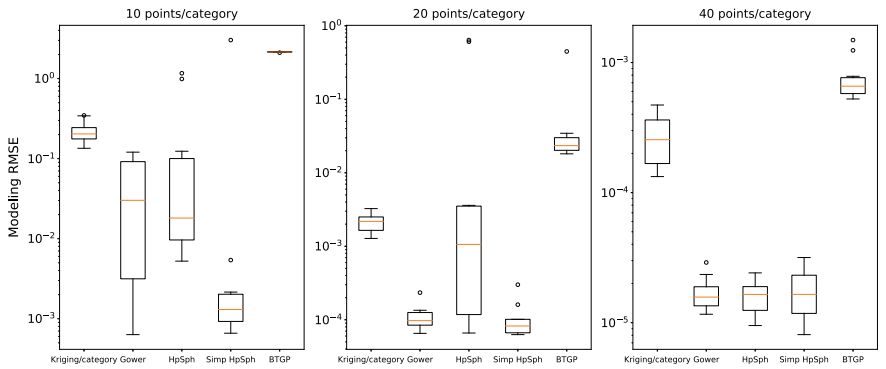


Fig. 6 Modeling RMSE for the Goldstein function obtained on 3 separate batches of 10 training data sets of varying size. From left to right, the used modeling methods are: a separate Kriging model per discrete category, a Gower distance based Gaussian process, a standard hypersphere decomposition based Gaussian process, a dimension-wise hypersphere decomposition based Gaussian process and a BTGP

The values of x_3 and x_4 are determined as a function of z_1 and z_2 according to the relations defined in Table 4.

The results obtained when modeling the Goldstein function with the various mixed continuous/discrete Gaussian processes are provided in Fig. 6. The testing is repeated with data sets of 90, 180 and 360 samples (i.e., 10, 20 and 40 samples per discrete category). It can be seen that when increasing the discrete size of the problem (in terms of number of categories), the standard hypersphere decomposition Gaussian process becomes slightly less performant than its dimension-wise variant, due to the increase in the number of hyperparameters that must be determined. More specifically, the difficulty in finding the optimal hyperparameters describing the standard hypersphere decomposition Gaussian process is highlighted by the relatively large standard deviation which characterizes the results obtained with said method. This trend is slightly mitigated for larger DoE, as enough data for the proper characterization of all the hyperparameters becomes available. It can also be noticed that for this test-case, the results obtained with the Gower distance Gaussian process are similar to the ones obtained with the standard hypersphere decomposition based ones, while the BTGP performs worse than the category-wise Kriging.

Table 5 Variables characterizing the combustion performance simulation test-case

Variable	Nature	Min	Max	Levels
O_F	Continuous	2.5	5.5	[–]
P_c [bar]	Continuous	20	80	[–]
ϵ	Continuous	10	60	[–]
Oxidant	Discrete	[–]	[–]	O2, N2O4, F2, H2O2
Reductant	Discrete	[–]	[–]	CH4, N2H4, JP-4, H2

4.4 Propulsion Performance Simulation

In order to better assess the surrogate modeling performance of the mixed continuous/discrete Gaussian processes described in the previous section, a number of aerospace design related test-cases are used as benchmarks. The first representative test-case is a simulation of the combustion performance of a launcher engine. More specifically, the modeled variable is the specific impulse I_{SP} provided by the engine. The I_{SP} is modeled as a function of the reductant to oxidant ratio O_F , the combustion chamber pressure P_c , the nozzle area ratio ϵ (defined as the ratio between the nozzle throat diameter and the nozzle exit diameter) and the type of reductant and oxidant. The first three variables characterizing the I_{SP} are continuous, while the type of reductant and oxidant are discrete choices which can be represented with the use of discrete variables. A summary of the variables characterizing the problem is provided in Table 5.

It is important to note that not all the combinations of reductant and oxidant can realistically be simulated and therefore only 7 possibilities were modeled. Due to the fact that not all the categories of the problem are included in the training data set, the standard hypersphere decomposition Gaussian process is not applied to this test-case. The combustion simulations necessary to create the training data sets are performed by using the thermo-chemical simulation software ‘Chemical Equilibrium with Applications’ (CEA) created by NASA [25]. This software simulates the combustion of gases in a combustion chamber and their expansion in an engine nozzle. The conditions for chemical motion equilibrium are stated in terms of Gibbs and Helmholtz energies [21] or the maximization of the entropy. The system of equations which characterizes the equilibrium and describes its composition is non-linear and therefore iterative methods (such as the Newton-Raphson method [6]) are used to solve it. In Fig. 7, the I_{SP} profiles for the 7 considered combinations of reductant and oxidant are shown as a function of the nozzle ratio ϵ and the reductant to oxidant ratio O_F .

The results obtained when modeling the launcher engine specific impulse are provided in Fig. 8. The testing is repeated with data sets of 56, 105 and 140 samples (i.e., 8, 15 and 20 samples per discrete category). Similarly to the analytical test-cases, the use of both Gower distance Gaussian processes and hypersphere decomposition ones considerably improves the surrogate modeling accuracy when compared to the category-wise Kriging. However, due to the fairly smooth nature of the response,

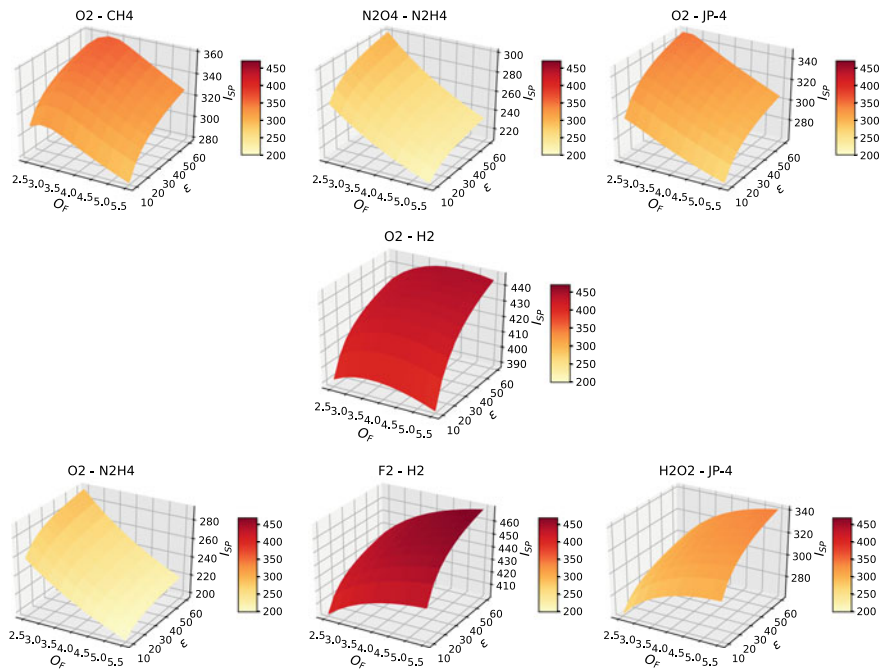


Fig. 7 Specific impulse of a launcher engine as a function of the nozzle ratio ϵ and the reductant to oxidant ratio O_F for various combinations of oxidant and reductant

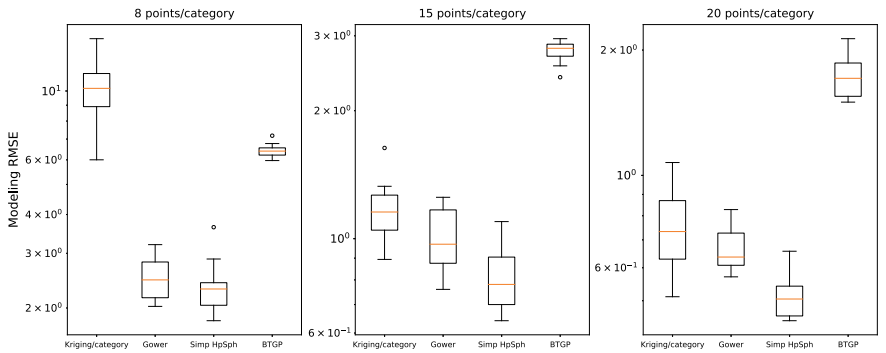


Fig. 8 Modeling RMSE [s] for the CEA combustion simulation obtained on 3 separate batches of 10 training data sets of varying size. From left to right, the used modeling methods are: a separate Kriging model per discrete category, a Gower distance based Gaussian process, a dimension-wise hypersphere decomposition based Gaussian process and a BTGP

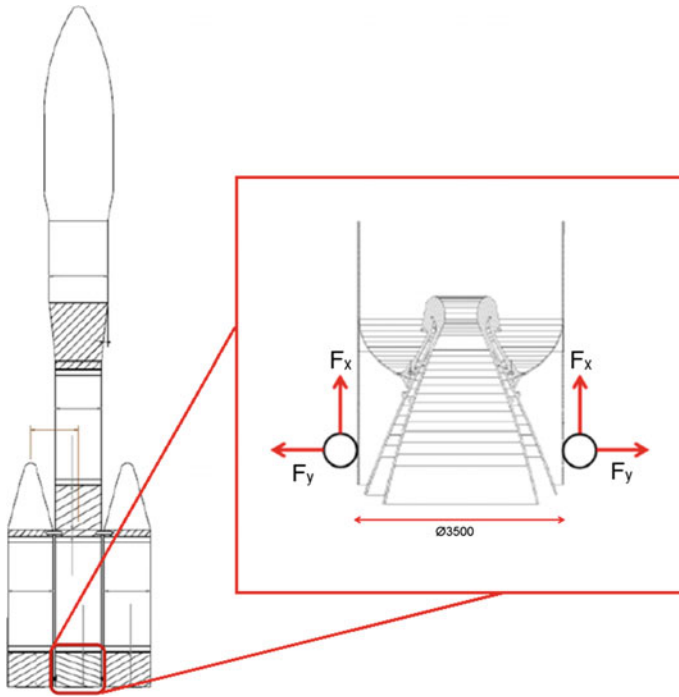


Fig. 9 Ariane 6 PPH launcher aft bay

the difference in modeling performance is not as pronounced as for the analytical test-cases. Similarly to Goldstein function test-case, the dimension-wise hypersphere decomposition Gaussian process outperforms the other surrogate models for all three training data set sizes that are considered.

4.5 Thrust Frame Structural Analysis

A second aerospace design test-case that is considered is the modeling of a launcher thrust frame stiffening, commonly performed for preliminary sizing purposes. More specifically, the Ariane 6 aft bay structural characteristics are analyzed. The thrust frame is located at the bottom of the launcher first stage and has the purpose of withstanding the weight of the launcher as well as the thrust of the two solid rocket boosters during the lift-off phase. A schematic representation of the Ariane 6 aft bay and its location within the launch vehicle system is provided in Fig. 9. This structure is composed of a cylindrical outer skin, stiffened by frames and stringers, and an inner body comprising three major frames and an inner skin. For illustrative purposes, two of the possible thrust frame configurations are illustrated in Fig. 10.

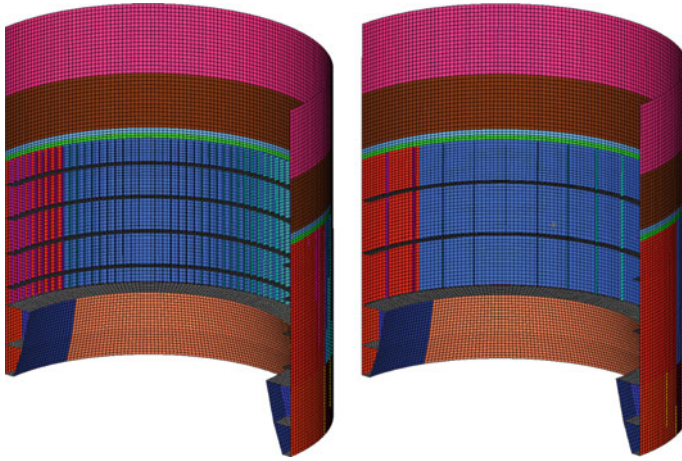


Fig. 10 Two examples of possible thrust frame configurations. On the left figure 144 stringers and 4 frames are included while on the right one only 36 stringers and 2 frames are present

The static loads that are considered for this simulation are the longitudinal and lateral thrust of the two solid rocket boosters (see Fig. 9). The boundary conditions are defined by modeling the first stage composite bottom skirt clamped to the upper interface of the thrust frame. In this test-case, two different parameters of the considered system are modeled: the maximum Von Mises stress on the inner skin of the structure [12] and the upper interface longitudinal over-flux. This choice is due to the fact that the same simulation can be used in order to compute both stiffness parameters with no additional cost. These parameters are modeled as a function of the inner and outer skins thicknesses t_i , t_o of the 6 regions in which the thrust frame is divided as well as of the number of stringers N_s and the number of frames N_f . The 12 thicknesses are continuous variables while the number of stringers and frames are discrete variables, each one characterized by 3 levels, thus resulting in a total of 9 categories. For illustrative purposes, structural responses on the entire thrust frame structure considering the maximum Von Mises stress and the over-flux are provided in Fig. 11. For the sake of simplicity, the same range is considered for all the thicknesses variables characterizing the problem. More specifically, the minimum and maximum bounds are 1 and 30 mm, respectively. A summary of the variables characterizing the studied problem is provided in Table 6.

In order to generate the training and testing data sets for this analysis, the MSC Nastran Finite Element Method (FEM) software [24] is used. In practice, a separate finite element model is created for every considered category of the problem (i.e., for every combination of the number of stringers and frames) due to the need of a distinct meshing for every configuration. A number of static FEM analyses is then performed with varying inner and outer skins thicknesses, according to the values present in the data sets, and using the finite element model corresponding to the category the considered sample belongs to. The lift-off conditions are simulated by considering

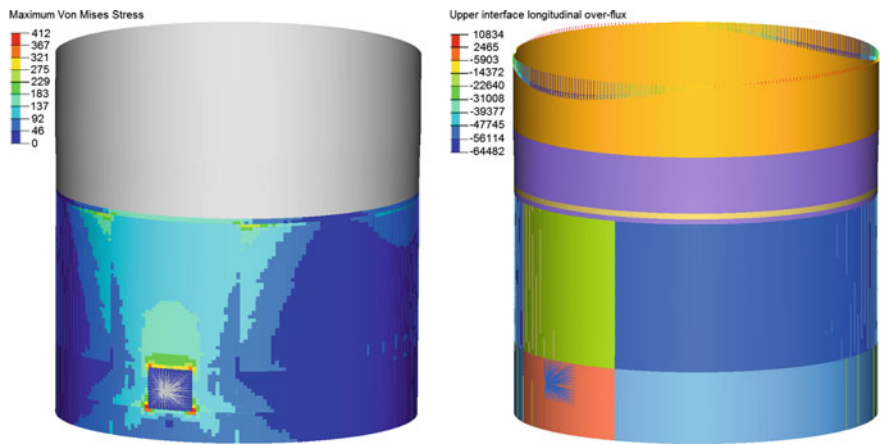


Fig. 11 Examples of structural responses on the entire thrust frame structure. On the left figure the maximum Von Mises stress is illustrated, while on the right figure the upper interface longitudinal over-flux is shown

Table 6 Variables characterizing the thrust frame structural analysis test-case

Variable	Nature	Min	Max	Levels
$t_{i,6}$ (mm)	Continuous	1	30	[–]
$t_{o,6}$ (mm)	Continuous	1	30	[–]
N_s	Discrete	[–]	[–]	36, 72, 144
N_f	Discrete	[–]	[–]	2, 4, 8

Table 7 RMSE [MPa] obtained for the surrogate modeling of the maximum inner skin Von Mises stress on a launcher thrust frame in a static thrust case

	Kriging/category	Gower	Hypersphere	‘Simp’ hypersphere	BTGP
DoE 1	322.45	263.08	296.90	316.76	296.91
DoE 2	325.01	284.91	291.29	283.45	338.83

two aligned vertical loads F_x and two opposing horizontal loads F_y , applied on the side of the thrust frame, as illustrated in Fig. 9.

The results obtained when performing the surrogate modeling of the maximum inner skin Von Mises stress and the upper interface longitudinal over-flux on the thrust frame are shown in Tables 7 and 8, respectively. The testing is repeated for 2 training data sets of 540 samples (i.e., 60 samples per discrete category in a 12-dimensional continuous search space). From these results it can be noted that for this test-case as well, mixed continuous/discrete surrogate modeling techniques produce overall better results than a standard category-wise Kriging, especially for what concerns the modeling of the upper interface longitudinal over-flux. Differently from the previous test-cases, it is the Gower distance based Gaussian process which on average produces the best results, while the two hypersphere decomposition based Gaussian processes

Table 8 RMSE [N] obtained for the surrogate modeling of the upper interface longitudinal over-flux on a launcher thrust frame in a static thrust case

	Kriging/category	Gower	Hypersphere	‘Simp’ hypersphere	BTGP
DoE 1	24484,72	12696,94	14375,19	14025,86	11129.44
DoE 2	21441,34	11055,39	14591,08	14785,67	14023.53

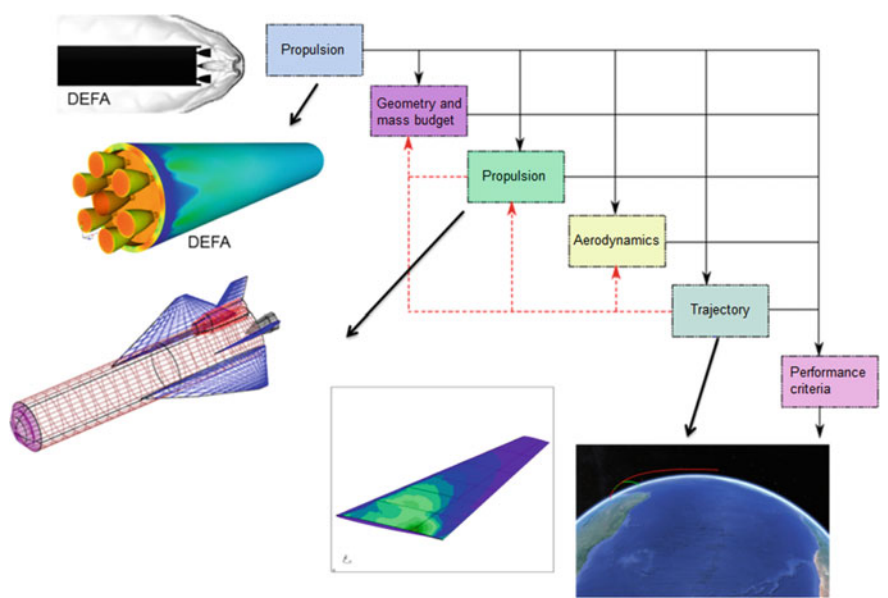


Fig. 12 Example of MultiDisciplinary Analysis for launch vehicles

result in a slightly worse surrogate modeling accuracy. It can also be noted that the performance of the BTGP varies considerably as a function of the DoE. Most notably, when considering the surrogate modeling of the over-flux, the BTGP produces the most accurate results with one of the two training data sets, but noticeably worse results with the other one.

4.6 Re-Usable Launch Vehicle System Flight Simulation

The last aerospace design test-case considered in this chapter is the multidisciplinary simulation of the flight of a re-usable launch vehicle as a function of a number of continuous and discrete design variables. The design of launch vehicles involves several disciplines and is customarily decomposed into interacting sub-models for propulsion, aerodynamics, trajectory, mass and structure, as is shown schematically in Fig. 12. The launch vehicle performance estimation which results from flight perfor-

Table 9 Re-usable launcher design variables

Variable	Nature	Min	Max	Levels
M (kg)	Continuous	20000	40000	[–]
A (m ²)	Continuous	20	30	[–]
I_{SP} (s)	Continuous	310	325	[–]
N_b	Discrete	[–]	[–]	2, 3, 4
Wing type	Discrete	[–]	[–]	type 1, type 2, type 3
Fairing type	Discrete	[–]	[–]	type 1, type 2

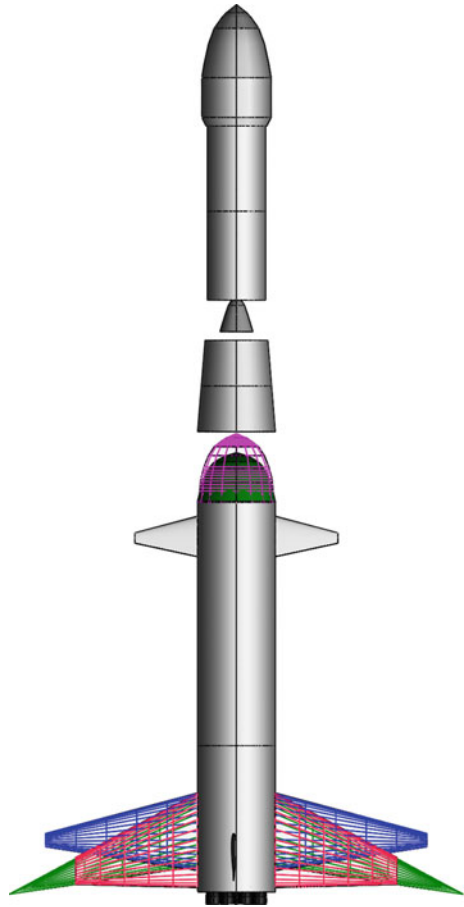
mance, reliability and cost, requires coupled disciplinary analyses [2]. The different disciplines are a primary source of trade-offs due to the opposing disciplinary effects on the launcher performance.

The considered test-case consists of a re-usable launch vehicle analysis problem in which the first stage returns to the launch site in glider mode with the help of its lifting surfaces (e.g., wings). The modeled variable is the distance between the launch site and the landing point as a function of the design variables. More specifically, the 6 input variables included in the test-case are the propellant mass M , the wing area A , the engine specific impulse I_{SP} , the number of engines used for the return maneuver boost N_b , the wing type and the fairing type (i.e., nose shape). The first 3 design variables are continuous while the latter 3 are discrete variables which characterize 18 distinct categories. Please note that in an actual design project, the landing distance offset should tend to zero for the launcher to be considered re-usable. A summary of the design variables characterizing the launcher test-case is provided in Table 9 and for illustrative purposes, a representation of the considered options for the wing and fairing types is shown in Fig. 13.

The launch vehicle analysis involves different disciplines: propulsion, sizing, aerodynamics and trajectory. The coupling between these disciplines is handled with the help of the multidisciplinary optimization platform OpenMDAO [15]. A brief detail on aerodynamics and trajectory disciplines is given in the following paragraphs while sizing and mass estimation are performed using the models presented in [7].

The aerodynamics discipline consists in determining the aerodynamics coefficients, such as the drag and lift coefficients required to compute the aerodynamics loads during the launch vehicle atmospheric flight. The calculations of the drag and lift coefficients are based on an ONERA code named MISSILE [8], which relies on an experimental database to determine the aerodynamics forces and coefficients of complex rocket geometries. For the trajectory estimation, a three-dimensional model with a rotating, round Earth is used. The trajectory control sequence (illustrated in Fig. 14) for the return is composed of a separation with the 2nd stage, followed by a turnaround maneuver and a rocket engine boost phase, where the number of engines to be used is one of the design variables. Subsequently, after reaching the apogee, the launch vehicle stage re-enters into the atmosphere at a zero angle of attack followed by an aerodynamic pull-up phase and glider return.

Fig. 13 Alternative options for the fairing and the wings of the re-usable launcher



The ODE dynamics equations are numerically integrated with a 4th order Runge Kutta solver [38]. The trajectory discipline consists of an optimization problem in which the optimal control is performed with a direct method [4]. The objective of trajectory optimization is to minimize the distance between the landing point and the landing site. The trajectory optimization consists in defining crossing points for the pitch angle ϑ (trajectory command law) and modifying them in order to satisfy the mission requirements. The trajectory discipline computes the loads (axial load factor, dynamic pressure) required to simulate the mass budget and geometry design discipline.

The results obtained when performing the surrogate modeling of the landing distance from the launch site are shown in Table 10. The testing is repeated for 2 training data sets of 504 samples (i.e., 28 samples per discrete category). As could be expected from the previous results, the large number of categories characterizing the problem results problematic for the convergence of the hypersphere decomposition hyperparameter optimization, due to the fact that a very large number of hyperparameters

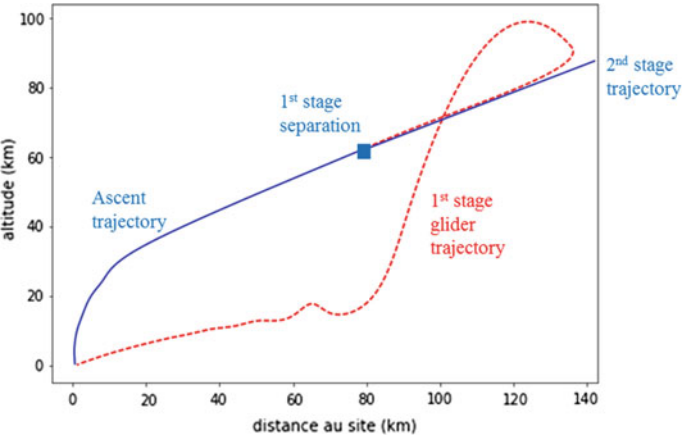


Fig. 14 Glider trajectory sequence

Table 10 RMSE [km] obtained for the surrogate modeling of the re-usable launch vehicle landing distance from launch site with different Gaussian processes

	Kriging/category	Gower	Hypersphere	‘Simp’ hypersphere	BTGP
DoE 1	9.353	4.285	4.567	4.227	8.346
DoE 2	6.819	3.554	4.629	3.532	6.708

Table 11 Number of hyperparameters characterizing the Gaussian processes for the re-usable launcher test-case

	Kriging/category	Gower	Hypersphere	‘Simp’ hypersphere	BTGP
N°	108	12	159	14	Varying

must be determined. By consequence, for this test-case the dimension-wise (‘simplified’) variant of the hypersphere decomposition Gaussian process produces more accurate surrogate modeling results than the other surrogate modeling techniques. For illustrative purposes, the number of hyperparameters that must be determined for the characterization of the various Gaussian processes in the re-usable launcher test-case are shown in Table 11. It can also be noticed that the Gower distance Gaussian process produces considerably accurate results for this particular test-case, which might be explained by the absence of anti-correlation trends between different values of the same discrete variable.

5 Results Synthesis and Analysis

In the previous section, several Gaussian process based surrogate modeling techniques for mixed continuous/discrete functions are compared in terms of modeling accuracy on a number of test-cases. The performance assessment on these test-cases is performed on data sets of varying size in order to show the dependency of the modeling accuracy on the size of the training data set. It is shown that among the presented surrogate models, the 2 hypersphere decomposition based Gaussian processes provide a considerably better modeling accuracy when compared to a category-wise Kriging, which represents the industry standard for mixed variable modeling. A clear advantage in modeling the continuous and the discrete variables together rather than creating a separate continuous model for every discrete category of the problem is therefore shown. For problems characterized by a small number of discrete categories, the surrogate modeling methods which produces the most accurate results is the standard hypersphere decomposition Gaussian process. However, when the combinatorial size of the considered problem increases, the number of hyperparameters that must be optimized during the training of the standard hypersphere decomposition Gaussian process becomes too large with respect to the available data, thus resulting in a surrogate modeling accuracy loss. In this cases, the dimension-wise variant of the hypersphere decomposition Gaussian process results more performant. For complex problems with large number of continuous and discrete variables, such as aerospace systems design, it is reasonable to assume that the dimension-wise variant of the hypersphere decomposition Gaussian process is a relevant choice due to the better scaling with the discrete dimension of the problem as well as its ability to model discrete categories which are not instantiated in the training data set.

6 Conclusions and Perspectives

In this chapter, a review of existing Gaussian process based surrogate modeling techniques for mixed continuous/discrete functions is provided. The presented surrogate modeling techniques are described and compared both from an analytical perspective and in terms of modeling accuracy on a number of test-cases. The dependency of the surrogate modeling performance of the various techniques on the continuous and discrete dimensions of the considered function as well as on the number of categories that characterize it is also discussed and shown with the help of the different test-cases. Although the various Gaussian processes produce a varying surrogate modeling accuracy as a function of the considered test-case and of the size of the training data set, it is shown that except for a few out-layers, the mixed variable surrogate modeling methods provide considerably better results than what can be obtained when modeling separately every category of the problem, which is the industry standard for mixed continuous/discrete functions surrogate modeling.

Among the presented alternatives to the category-wise Kriging for the surrogate modeling of mixed variable functions, the dummy coding based BTGP is the one that resulted in the lowest surrogate modeling accuracy over the ensemble of test-cases. However, it is the opinion of the authors that a partition tree based approach similar to the one adopted by the BTGP could provide an answer to a number of mixed continuous/discrete surrogate modeling challenges, such as the complexity of including in the same model variables characterizing considerably different behaviors as well as the scaling with the dimension of the problem. If the dummy coding is replaced with a more efficient and refined method to handle the discrete variables, both during the optimal partition tree definition process and in the local surrogate modeling, the resulting tree based Gaussian process could potentially present better performance and better scaling than the techniques described in this chapter for the modeling of complex mixed continuous/discrete functions.

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